RiskQuant · Credit VaR

Measuring the losses that matter when counterparties fail together

EXECUTIVE SUMMARY

Part 0 · Where We Stand Today

What Credit VaR needs, what exists today, and what can run now

> ***The foundation is already built. The remaining question is how far we want to extend it.***

What Credit VaR Needs

Four inputs. The first two size each loss; the last two decide how often losses land, and how badly they cluster.

#	Input	In plain terms
1	**Exposure**	How much value is at risk when a counterparty defaults
2	**Loss severity (LGD)**	How much of that we would actually lose after recovery
3	**Credit quality (PD)**	How likely each counterparty is to default
4	**Default dependence**	How likely they are to default *together*

> ***The first three determine expected loss. The fourth determines how severe the tail can become.***

What We Already Have

Of the four inputs, three are in hand today. One is a separate build.

#	Input	Status
1	Exposure (EAD)	**Gap** · current MTM works today; a forward-looking exposure engine is a separate investment
2	Loss severity (LGD)	**Have** · recovery and credit assumptions
3	Credit quality (PD)	**Have** · ratings, market signals, credit data
4	Default dependence	**Have** · the *RiskQuant* model suite is already built

> ***We can run today using current and stressed MTM as exposure. A proper counterparty-exposure engine (future exposure over time) is the one piece worth investing in.***

What We Can Do Today

Run the models now on the live OTC book, at current MTM and a stressed MTM, and read each loss as a share of total unrealised MTM.

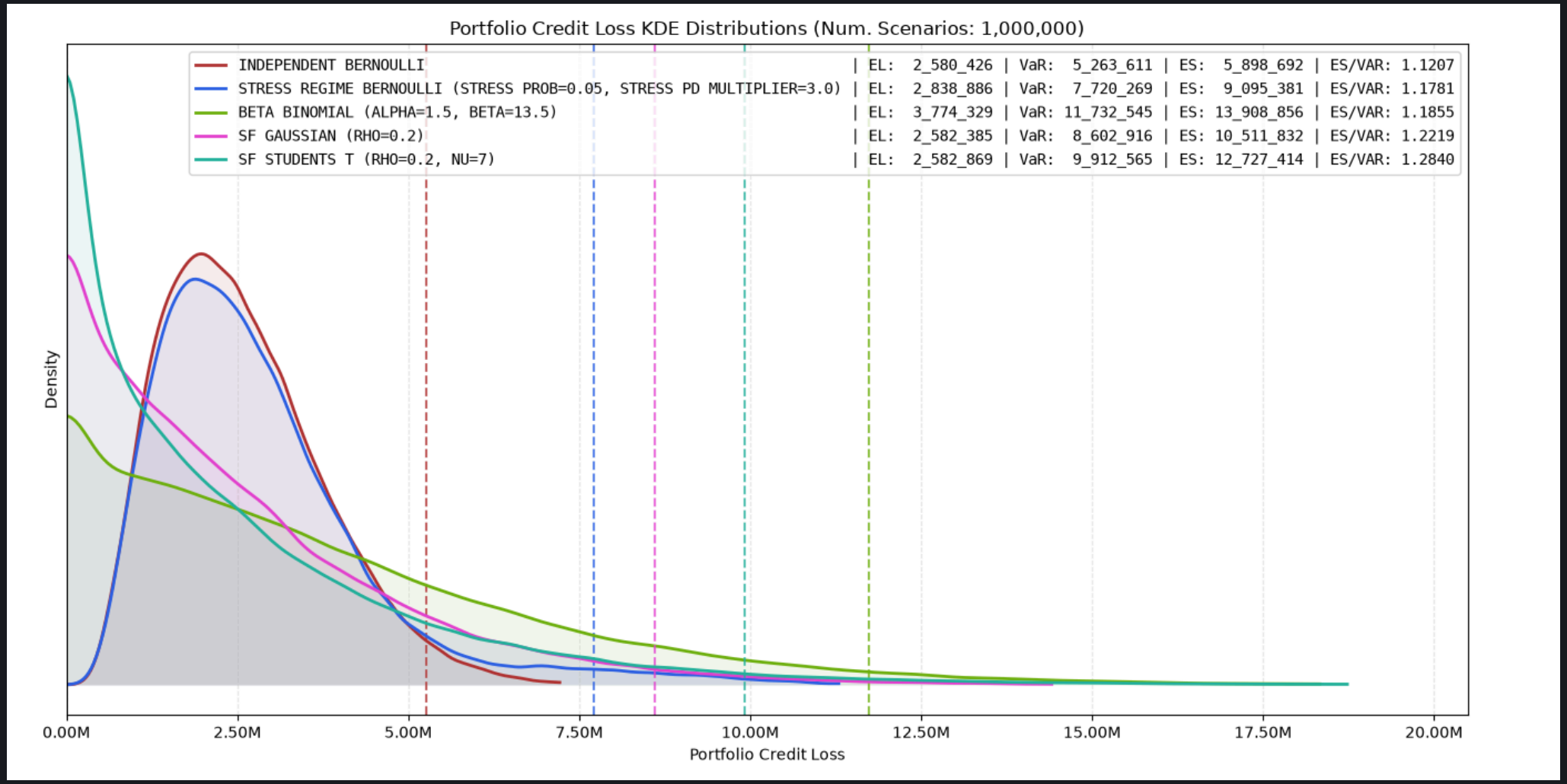
Example: STX B.V. · GOO portfolio (full charts in the appendix)

Model	CrVaR @ MtM	CrVaR @ MtM +20%	CrVaR @ MtM -20%	What it assumes
Independent Bernoulli	5,263,611	9,024,797	8,003,460	Defaults occur independently
Stress-Regime Bernoulli	7,720,269	13,694,149	9,854,062	Defaults increase together in defined stressed states
Beta-Binomial	11,732,545	15,329,989	14,485,760	Portfolio default probability is uncertain
Single-Factor Gaussian Copula	8,602,916	14,008,345	11,284,892	Defaults driven by a shared factor (e.g., global credit)
Single-Factor Student-t Copula	9,912,565	15,612,723	12,582,565	Shared factor with stronger tail clustering

> ***Same portfolio. Similar expected loss. The difference is the tail: how each model believes defaults can cluster.***

** Actual OTC positions and market values used. Credit assumptions (PD/LGD) are illustrative inputs. Stress cases revalue the portfolio under EUR price shocks and recompute exposure. No collateral benefit assumed.*

STX B.V. - GOO - Base Case



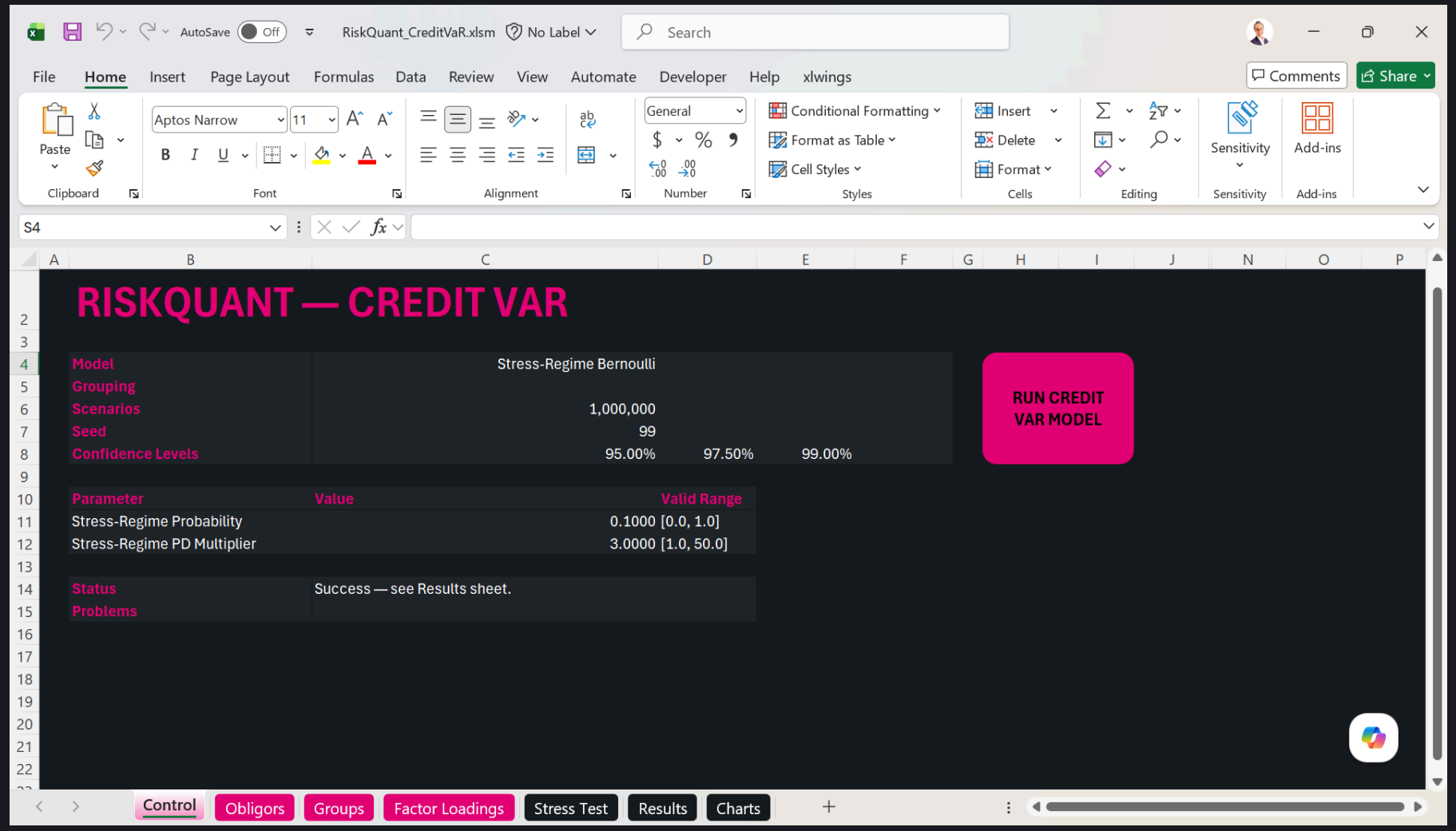
Credit VaR Is Operational Today

Credit VaR is already available through an Excel interface for *immediate* use by the credit and trading teams.

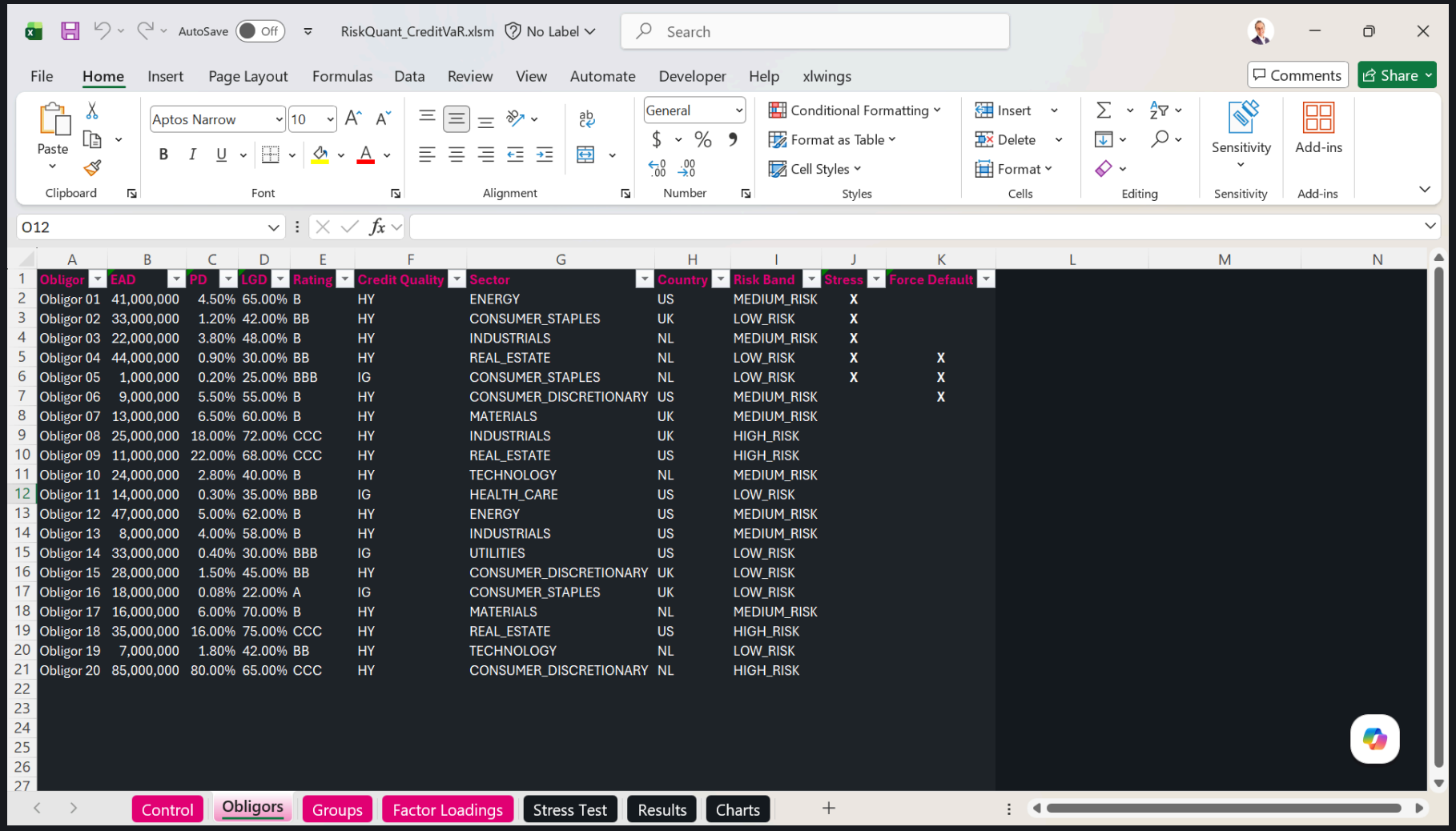
- **Configure:** enter counterparties, EAD, PD, LGD, and select the dependence model
- **Run:** generate loss distributions, VaR, ES, and capital metrics in one click
- **Understand:** identify the counterparties driving tail risk through ES contribution and incremental risk
- **Stress:** force defaults or modify exposures externally and immediately re-run scenarios
- **Today:** current/stressed MTM-based counterparty EAD measurement
- **Next step:** integrate forward-looking CCR exposure profiles for full EAD modelling

> ***The engine exists. The analytics exist. The decision framework exists.***

RiskQuant Credit VaR Excel Tool



RiskQuant Credit VaR Excel Tool



Obligor	EAD	PD	LGD	Rating	Credit Quality	Sector	Country	Risk Band	Stress	Force Default
Obligor 01	41,000,000	4.50%	65.00%	B	HY	ENERGY	US	MEDIUM_RISK	X	
Obligor 02	33,000,000	1.20%	42.00%	BB	HY	CONSUMER_STAPLES	UK	LOW_RISK	X	
Obligor 03	22,000,000	3.80%	48.00%	B	HY	INDUSTRIALS	NL	MEDIUM_RISK	X	
Obligor 04	44,000,000	0.90%	30.00%	BB	HY	REAL_ESTATE	NL	LOW_RISK	X	X
Obligor 05	1,000,000	0.20%	25.00%	BBB	IG	CONSUMER_STAPLES	NL	LOW_RISK	X	X
Obligor 06	9,000,000	5.50%	55.00%	B	HY	CONSUMER_DISCRETIONARY	US	MEDIUM_RISK		X
Obligor 07	13,000,000	6.50%	60.00%	B	HY	MATERIALS	UK	MEDIUM_RISK		
Obligor 08	25,000,000	18.00%	72.00%	CCC	HY	INDUSTRIALS	UK	HIGH_RISK		
Obligor 09	11,000,000	22.00%	68.00%	CCC	HY	REAL_ESTATE	US	HIGH_RISK		
Obligor 10	24,000,000	2.80%	40.00%	B	HY	TECHNOLOGY	NL	MEDIUM_RISK		
Obligor 11	14,000,000	0.30%	35.00%	BBB	IG	HEALTH_CARE	US	LOW_RISK		
Obligor 12	47,000,000	5.00%	62.00%	B	HY	ENERGY	US	MEDIUM_RISK		
Obligor 13	8,000,000	4.00%	58.00%	B	HY	INDUSTRIALS	US	MEDIUM_RISK		
Obligor 14	33,000,000	0.40%	30.00%	BBB	IG	UTILITIES	US	LOW_RISK		
Obligor 15	28,000,000	1.50%	45.00%	BB	HY	CONSUMER_DISCRETIONARY	UK	LOW_RISK		
Obligor 16	18,000,000	0.08%	22.00%	A	IG	CONSUMER_STAPLES	UK	LOW_RISK		
Obligor 17	16,000,000	6.00%	70.00%	B	HY	MATERIALS	NL	MEDIUM_RISK		
Obligor 18	35,000,000	16.00%	75.00%	CCC	HY	REAL_ESTATE	US	HIGH_RISK		
Obligor 19	7,000,000	1.80%	42.00%	BB	HY	TECHNOLOGY	NL	LOW_RISK		
Obligor 20	85,000,000	80.00%	65.00%	CCC	HY	CONSUMER_DISCRETIONARY	NL	HIGH_RISK		

FORBIDDEN MATERIALS

RiskQuant · Credit VaR

The Details

Measuring the losses that matter when counterparties fail together

Part 1 · Credit VaR

What Credit VaR Measures

Credit VaR answers one question: *if OTC counterparties default, what losses could our trading portfolio experience?*

> ***Expected Loss tells us what we expect. Credit VaR measures what we must survive.***

Component	Question	Determines
Exposure	How much value is at risk if a counterparty defaults?	The size of each loss
Default probability	How likely is each counterparty to default?	The average loss
Default dependence	How likely are counterparties to default together?	**The extreme losses (tail)**

Why Credit VaR Exists

Expected Loss measures the ***average counterparty credit loss***.

Credit VaR measures the ***tail losses that drive internal capital and risk appetite***.

Expected Loss	Credit VaR
"What is our average credit loss?"	"What loss could we face in a severe event?"
Driven by exposure, PD and recovery	Driven by exposure, PD and **default dependence**
Measures the average impact of counterparty defaults	Measures the extreme outcomes that drive risk appetite and capital

> ***The key uncertainty is not only whether counterparties default, but whether they default together.***

The Business Questions Behind the Tail

- **How much** could we lose in a severe counterparty event?
- **Which counterparties** contribute most to that tail?
- **Where are we concentrated** by counterparty, and in the shared risk driver?
- **How much diversification do we *actually* have** versus assume?
- **How resilient are we** if several counterparties fail together?

> ***Credit risk is difficult not because defaults are mysterious, but because they can occur together. Every model in this deck represents a different view of that dependence.***

The Core Data Behind Credit VaR

Credit VaR combines four inputs.

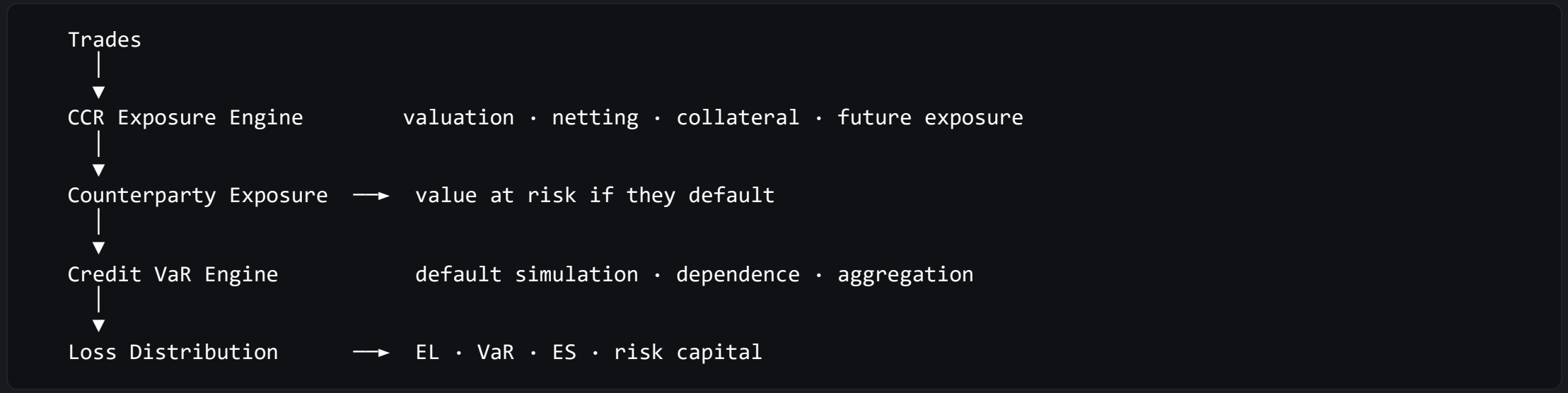
The first two determine the **size of a loss**.
The last two determine **how often losses occur and how severely they cluster**.

Input	Question answered	Source
Exposure (EAD)	How much value is exposed if a counterparty defaults?	CCR exposure engine
Loss severity (LGD)	How much of that exposure is lost after recovery?	Counterparty credit data
Counterparty credit quality (PD)	How likely is each counterparty to default?	Ratings, market signals, credit assumptions
Default dependence	How likely are defaults to occur together?	Model assumptions & calibration

> ***At the core, Credit VaR combines exposure, loss severity and default likelihood into a portfolio loss distribution. Dependence determines how much that loss grows in the tail.***

Where it Sits in the Risk Stack

- **CCR models** *size* exposure.
- **Credit VaR** turns exposure + default + dependence into a *loss distribution*.
- Credit risk comes from the possibility that a derivative has positive replacement value when the counterparty defaults.



Part 2 · Jump-to-Default (JTD)

From counterparty defaults to portfolio loss distributions

> ***A default creates a loss.***
> ***Default dependence determines how severe the portfolio tail can become.***

Jump-to-Default (In a Nutshell)

The event: a counterparty defaults, and the positive value of our OTC portfolio becomes a credit loss.

We don't lend principal. Our credit risk is the **OTC portfolio**: firms owing us mark-to-market on unrealised trades.

The loop, per scenario:

1. Exposure at default → 2. Does the name default? → 3. Apply LGD → 4. Sum across the portfolio → 5. Repeat millions of times → **portfolio loss distribution**

> ***The whole modelling choice reduces to steps 2 and 4: when one name defaults, how likely is the next?***

What JTD Credit VaR Does Not Answer

JTD Credit VaR measures the loss **when default happens**.

It does not answer:

Question	Owned by
How large could exposure become before default?	CCR modelling
What happens if markets gap?	Market risk
Does exposure increase as credit deteriorates?	Wrong-way risk analysis
How quickly can positions be closed?	Liquidity risk
How does credit quality deteriorate before default?	Credit migration models

> ***JTD Credit VaR isolates default loss. It is one component of total counterparty risk.***

The One Choice That Moves Capital

Driven by individual assumptions	Driven by dependence assumptions
Expected Loss	**The tail: VaR & ES**
Barely moves across models	Moves *enormously* across models

Example: €100m exposure, 40% recovery → **€60m loss if it defaults.**

The marginal models agree on how often that costs us on average. They disagree on how often it happens *alongside twenty others.*

> ***Get the exposure and default assumptions right for average losses. Get the dependence right for tail risk.***

The Same Portfolio. Different Views of the Tail.

Each model answers the same question: **When one counterparty defaults, how much does it change the likelihood that others default too?**

#	Model	Assumption added
1	Forced Default	Everything fails together
2	Independent Bernoulli	Defaults are unrelated
3	Beta-Binomial	Default rates can move higher or lower together
4	Stress-Regime Bernoulli	The market can enter a crisis state
5	Gaussian Copula	Counterparties share common risk drivers
6	Student-t Copula	Extreme tail dependence

> ***Same portfolio. Same Expected Loss. The difference is the tail.***

1 · Forced Default: *"assume everything fails all at once"*

Intuition: every counterparty defaults at once. No probability, no diversification.

Answers: *what is the absolute most we could lose?*

Strengths	Weaknesses
Needs almost no data	Not a real scenario
A hard, un-arguable ceiling	Not suitable as a realistic capital estimate

Use it for: a bounding sanity check, and brand-new (sub)portfolios with no history, hard stresses.

> ***The ceiling every other model must sit beneath.***

2 · Independent Bernoulli: *"unrelated coin flips"*

Intuition: each counterparty is a separate weighted coin flip. One failing tells you nothing about the next.

Answers: *what does the loss look like if defaults are unrelated?*

Strengths	Weaknesses
Simple, fast, full distribution	Forbids clustering entirely
Captures single-name concentration	**Understates the tail; flatters diversification**

Use it for: granular, homogeneous (sub)portfolios; the baseline everything else is measured against.

> ***Its diversification benefit is overstated because counterparties share common drivers.***

3 · Beta-Binomial: *"the default rate can move"*

Intuition: each counterparty has a default probability, but the portfolio default rate is not fixed. It moves from scenario to scenario.

Answers: *what happens when default probabilities are more uncertain than a fixed PD assumption implies?*

Strengths	Weaknesses
Introduces volatility in portfolio default rates	Treats the portfolio as one homogeneous pool
Creates default clustering without specifying risk factors	No sector / geography / counterparty-specific drivers

Use it for: portfolios where default uncertainty matters, but a full factor model is not required.

> ***The first model to admit that default probabilities move. That uncertainty alone creates clustering.***

4 · Stress-Regime Bernoulli: *"dependence through scenarios"*

Intuition: counterparties can move together because the portfolio can enter different states of the world. In a stressed state, default probabilities increase across names.

Answers: *what happens when multiple counterparties become more likely to default under a defined stress scenario?*

Strengths	Weaknesses
Makes default dependence explicit and interpretable	Reality is not always a small number of discrete states
Easy to link losses to defined stress scenarios	Requires judgement on stress severity and likelihood

Use it for: stress testing, limit discussions, understanding portfolio behaviour under adverse scenarios.

> ***Dependence is introduced through explicit stressed states.***

5 · Gaussian Copula: *the industry standard*

Intuition: each name is part idiosyncratic, part tied to a **shared economy.** When the economy dips, names weaken together and defaults cluster.

Answers: *how much diversification do we really have once counterparties share drivers?*

Strengths	Weaknesses
Market & regulatory standard	One economy; single factor
Realistic clustering & diversification	Bell-curve tails understate joint crashes

Use it for: portfolio tail risk measurement and counterparty risk capital.

> ***The workhorse, and the base the final model improves on.***

6 · Student-t Copula: *"extremes cluster"*

Intuition: counterparties share common risk drivers, but extreme market conditions create stronger co-movement than a Gaussian model allows.

Answers: *how severe can portfolio losses become when defaults cluster in the tail?*

Strengths	Weaknesses
Captures stronger tail dependence	Additional parameters require calibration
More realistic extreme-event behaviour	Can produce more conservative tail estimates

Use it for: understanding extreme portfolio losses and stress scenarios where diversification breaks down.

> ***In normal conditions counterparties may diversify. In extreme conditions, that diversification can disappear.***

Comparing the Models: No Single Model Wins Everywhere

Model	Realism	Transparency	Tail honesty	Data need	Business fit
Forced Default	–	High	–	Minimal	Stress bound
Independent Bernoulli	Low	High	Low	Low	Baseline view
Beta-Binomial	Med	High	Med	Low	Portfolio stress
Stress-Regime	Med	High	Med-High	Med	Scenario analysis
Gaussian Copula	High	Med	Med	Med-High	Core risk view
Student-t Copula	**High**	Med	**High**	Med-High	Extreme tail view

> ***Transparency and realism pull in opposite directions. Simpler models are easier to explain; richer models capture more complex dependence. That is why you need a suite.***

Which Model Answers Which Question?

The question we need to answer	Model
What is the absolute loss ceiling?	Forced Default
What if defaults are independent?	Independent Bernoulli
What if the portfolio default rate is higher or lower than expected?	Beta-Binomial
What happens in a defined stress scenario?	Stress-Regime Bernoulli
What if counterparties are exposed to the same economic drivers?	Gaussian Copula
What if extreme events create unusually strong clustering?	Student-t Copula

> ***The objective is not to find the "right" model. It is to understand how much of the tail is created by our assumptions about dependence.***

Part 3 · Reading the Output

From loss distribution to risk decisions

> ***The loss number is only the beginning. The real value is understanding what drives the tail and where risk comes from.***

The Metrics Every Model Returns

The same battery on every model's loss distribution. Models agree near the middle and separate in the tail.

Metric	Reads as	Feeds
Expected Loss (EL)	average counterparty default loss	expected credit cost
Loss volatility (σ)	how dispersed outcomes are	first tail signal
VaR	the loss we won't breach *x%* of the time	risk-appetite limits
Expected Shortfall (ES)	the *average* loss once we're past VaR	internal capital

> ***VaR is a line: a single point on the curve. ES is the territory beyond it: the average of everything worse.***

Across models, Expected Loss (EL) stays almost unchanged. *The difference appears in the tail.*

Incremental VaR & ES: *"what does this counterparty add?"*

Hold the portfolio constant, remove one counterparty, and re-measure the tail. The difference is its incremental risk.

```
incremental_risk(i) = risk(portfolio) − risk(portfolio excl. i)
```

Answers: *How much risk budget do we gain or lose by adding or removing a counterparty?*

Property	Consequence
Directional, drop-one / add-one	The **new-trade and limit-headroom** metric
Defined for both VaR and ES	Same methodology across risk measures
Not additive	Individual contributions do not sum to portfolio risk

Use it for: pre-trade decisions, counterparty limits, novation, and unwind analysis.

Contribution ES

A different question from incremental: not "what if we drop it," but *how is today's tail split across the portfolio*: an additive allocation that sums **exactly** to portfolio ES.

> `contribution_ES(i)` = ***average loss of*** `i` ***across the worst*** `1-α` ***scenarios.***

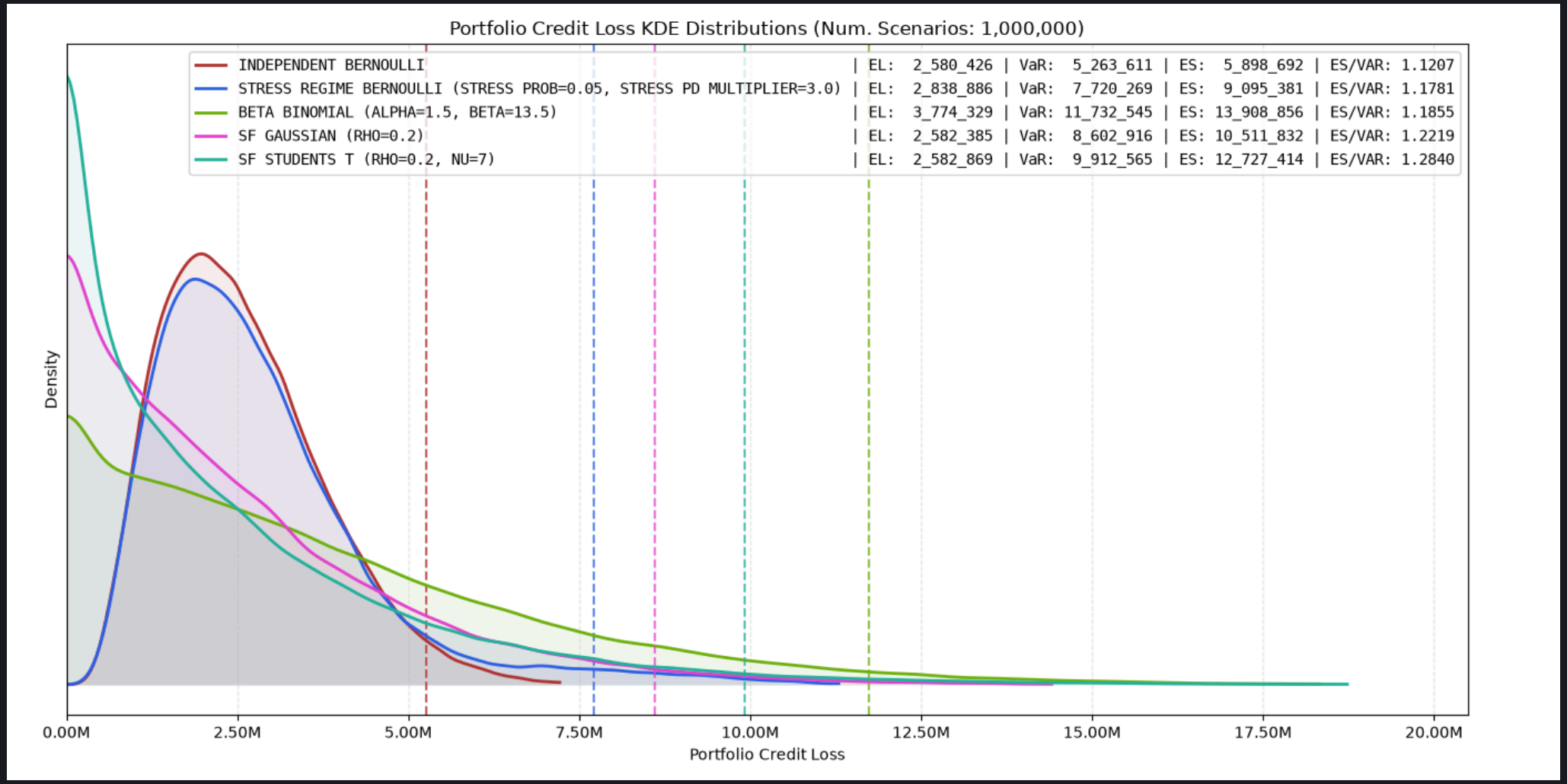
> `portfolio_ES = sum_i contribution_ES(i)`

RiskQuant · Credit VaR

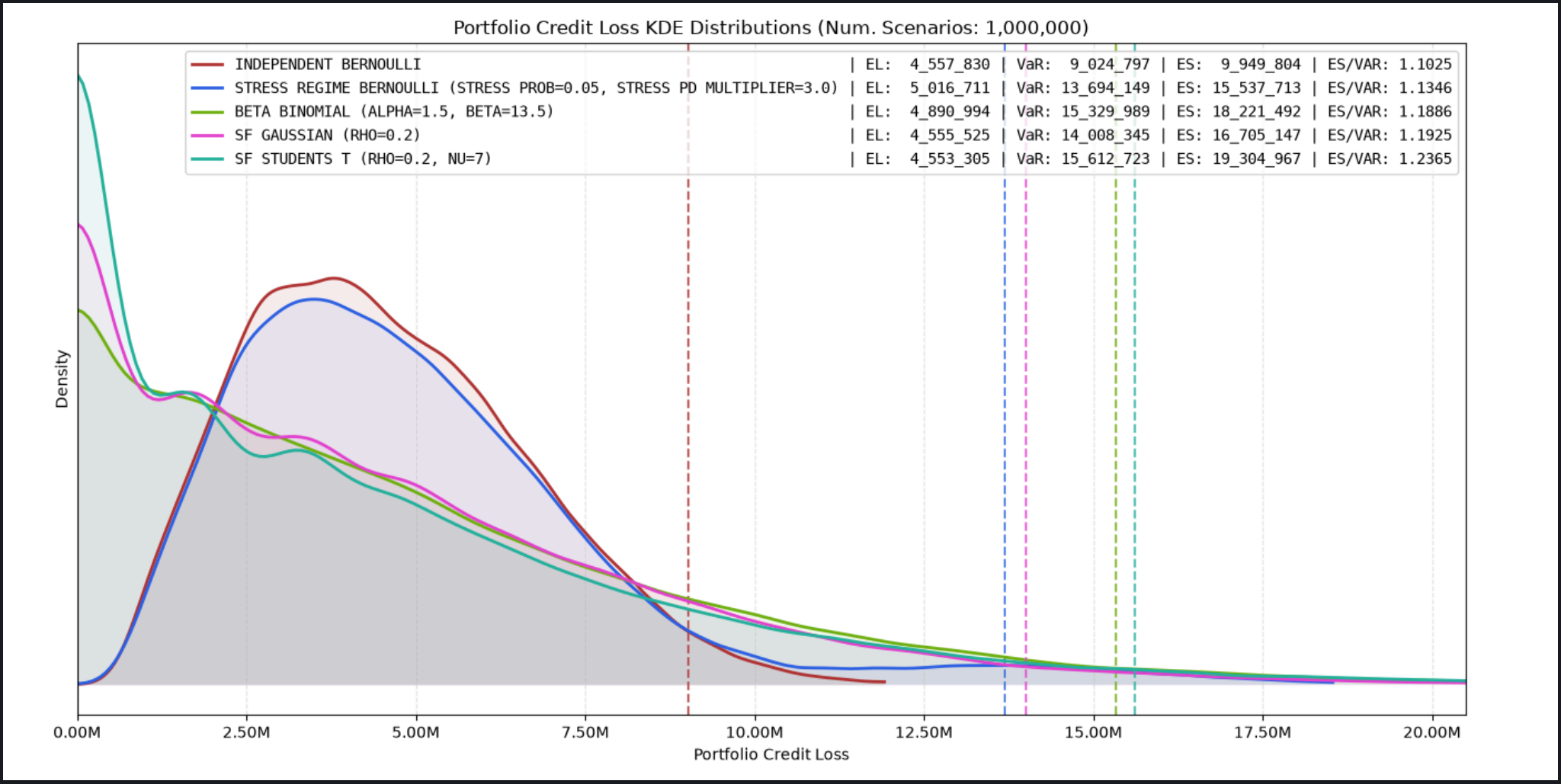
> *Defaults create losses. Dependence creates tail risk.*

Appendix: STX Credit VaR

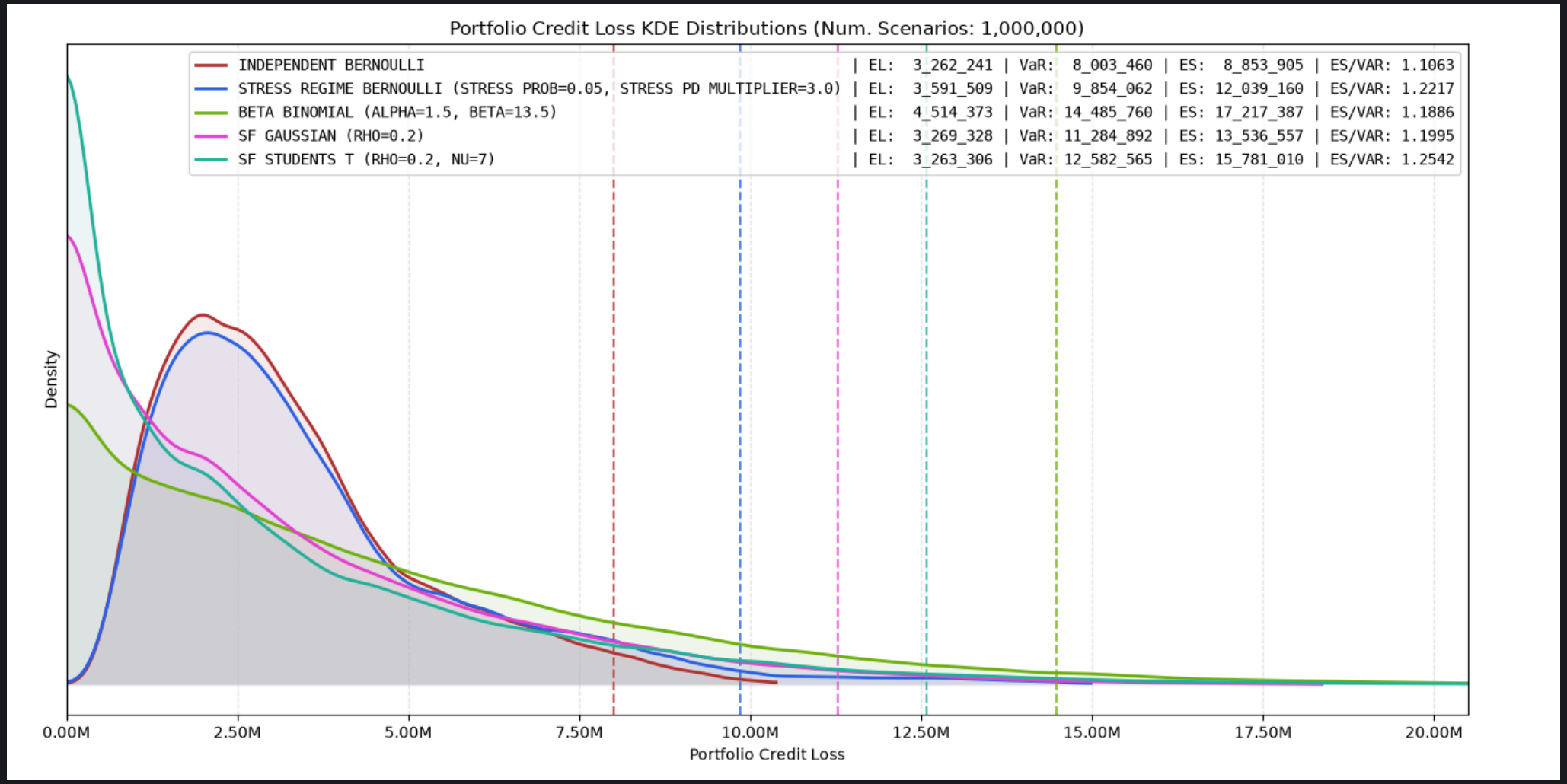
STX B.V. - GOO - Base Case



STX B.V. - GOO - Stressed (+20%)



STX B.V. - GOO - Stressed (-20%)



STX B.V. - GOO - Random Up/Down 10%-20% (PC Level)

